Supplementary Information
Deep Generative Model for Human Mobility Behavior

Ye Hong[a,*], Yatao Zhang[a,b], Konrad Schindler[c], Martin Raubal[a,b]

[a] *Institute of Cartography and Geoinformation, ETH Zurich*
[b] *Future Resilient Systems, Singapore-ETH Centre, ETH Zurich*
[c] *Photogrammetry and Remote Sensing, ETH Zurich*

Supplementary Note 1. Comparing microscopic mobility models

We compare the ability of several microscopic mobility models to capture the fundamental properties of individual location choices. The models include the exploration and preferential return (EPR) model [15], the container model [1], location prediction models such as the Markov model [6] and the multi-head self-attention (MHSA)-based neural network [9], as well as the deep generative MoveSim network [5]. Below, we provide a detailed description and implementation of each model.

- EPR model [15]. As a well-established microscopic mechanistic mobility model, EPR reproduces scaling laws for distinct location numbers and their visitation frequencies over time. We estimate the model parameters using the training set and determine, for each sequence in the test dataset, the number of visited locations S and the number of visits f_i for each location i. We then generate 50 displacement steps [15]: at each step, the individual chooses to either explore a new location with probability $p^{new} = \rho S^{-\gamma}$, where ρ and γ are model parameters, or return to a previously visited location with complementary probability $1-p^{new}$. During exploration, a new location is determined by sampling a moving distance Δr from the empirically estimated jump length distribution $P(\Delta r)$. When returning to a visited location, the probability of choosing location j, denoted as Π_j, is proportional to the number of previous visits to j, i.e., $\Pi_j \propto f_j$.

- Container model [1]. Alessandretti et al. [1] observed that individual mobility can be understood through nested spatial scales and introduced the container model to explain how individuals allocate time and move between these scales. The model features a hierarchical structure H with L levels, an attractiveness measure a for containers at each level, and an $L \times L$ matrix D containing displacement

*Corresponding author
Email addresses: hongy@ethz.ch (Ye Hong), yatzhang@ethz.ch (Yatao Zhang), schindler@ethz.ch (Konrad Schindler), mraubal@ethz.ch (Martin Raubal)

probabilities. The probability of traveling from the current location j to another location k is defined as:

$$p_{D_{j,h},D_{j,k}} \prod_{l \leq D_{j,k}} a(k_l)$$

where the first factor, $p_{D_{j,h},D_{j,k}}$, represents the probability of traveling a distance $D_{j,k}$, given that the individual is currently at distance $D_{j,h}$ from their home location h. The second factor denotes the probability of selecting location k at that level distance, where $a(k_l)$ represents the attractiveness of a container at level l that includes location k. We estimate the model parameters H, a, and D for each individual using the training set, following maximum likelihood estimation [1], and generate 50 displacements for test sequences.

- Markov model [6]. Classical location prediction models assume a Markov chain of individual location visits [2], treating locations as states and representing their pairwise transition probabilities in a Markov matrix. We implement a first-order Mobility Markov Chain (1-MMC) [6]. During inference, the next location visit for an individual is sampled from the top-3 most likely locations based on their transition probabilities from the current location. This process is repeated until the generated visit sequence reaches the desired length, which is 50 steps in this study.

- MHSA-based model [9]. State-of-the-art prediction models encode locations into an embedding space and learn their transition patterns using sequential deep learning networks [11]. We train an MHSA-based next-location prediction network and employ encoding strategies to autoregressively generate novel location sequences. At each generation step, we sample the next location using a combination of top-k and nucleus sampling strategies [8]. The top-k strategy limits the selection to the k most likely location predictions, while the nucleus sampling restricts the selection to locations whose cumulative probability exceeds a predefined threshold, p. We set $k = 200$ and $p = 0.99$. The predicted location is then appended to the input sequence and subsequently used in the next prediction. This process is repeated for 50 steps.

- MoveSim [5]. As a baseline for generative modeling of location choices, we implement MoveSim, a sequence-to-sequence generative adversarial network (GAN)-based model. MoveSim incorporates a generator network that simulates location sequences using mobility prior knowledge and a discriminator network that differentiates between real sequences and those generated by the generator. These two networks are trained iteratively using a min-max loss designed to optimize their performance jointly. We did not include historical transition, physical distance, or function similarity information as in the

original implementation, as each involves calculating a pairwise location matrix, and the number of unique locations in our dataset is extremely large.

We evaluate these models' performance using location metrics. In addition to the metrics described in the main text, we consider the jump length distribution [3] and the uncorrelated entropy [16]. The evaluation results are shown in Supplementary Figure S1. To assess model performance, we employed the log-likelihood ratio to compare the likelihood that the real data was generated by each model relative to MobilityGen. Additionally, we computed the Wasserstein distance between the metric distribution derived from the generated location visits and those from the real data (Extended Data Table 1 and Supplementary Table S1). Overall, MobilityGen provides a more accurate description of the data than other models across all metrics, except for jump length, where it slightly overemphasized longer displacements compared to the real data (Supplementary Figure S1d). Furthermore, the median evolution of the radius of gyration across individuals, $r_u(k)$, over displacement steps k, is well described by logarithmic growth (Supplementary Figure S1b). The fit $r_u(k) \propto \alpha \log(k)$, with parameters α derived from MobilityGen ($\alpha_{\text{MobilityGen}} = 2967 \pm 29$), aligns most closely with the real data ($\alpha_{\text{true}} = 2875 \pm 35$), while all other models have markedly different parameters ($\alpha_{\text{epr}} = 6431 \pm 185$, $\alpha_{\text{container}} = 3784 \pm 68$, $\alpha_{\text{markov}} = 796 \pm 26$, $\alpha_{\text{mhsa}} = 1541 \pm 12$, $\alpha_{\text{MovSim}} = 14976 \pm 462$).

Supplementary Note 2. Quality of generated activity locations

To complement the mobility metrics, we evaluate the simulated location sequences using metrics from language modeling, which quantify similarity and diversity with the real sequences. The similarity is evaluated using the Bilingual Evaluation Understudy (BLEU) score [12], an n-gram-based metric that compares each generated sequence with its corresponding ground truth, where higher values indicate greater similarity. We implement the sentence-level smoothed version from BLEU-1 to BLEU-4. For diversity, we use the distinct unigram (dist-1) metric [7] to measure intra-diversity, with a lower dist-1 indicating more repeated locations. For sequence-level diversity, we consider the diverse 4-gram (div-4) metric [4] to measure the ratio of distinct 4-grams in each sequence, where higher div-4 values suggest greater generation diversity. The evaluation results for all implemented microscopic mobility models are presented in Supplementary Table S2. MobilityGen can generate location visit patterns similar to the real data while avoiding input replication and maintaining a moderate level of diversity.

We note that these metrics should be interpreted with caution: (a) In language modeling tasks, multiple valid target sentences typically exist for evaluating BLEU, whereas the observed ground truth mobility

represents only one viable option within a broader set of possibilities. (b) Rather than solely encouraging diversity, the generation of locations should balance exploring new locations with returning to familiar ones. As such, diversity measures alone may not adequately reflect the realism of the generation.

Supplementary Note 3. Benchmarking generated activity-travel patterns

To evaluate the effectiveness of MobilityGen in reproducing spatiotemporal activity-travel patterns, we benchmark its performance against two state-of-the-art mobility frameworks: the Diary-based Trajectory Simulator (DITRAS) framework [13] and the TimeGeo framework [10]. Detailed descriptions of their methodologies and implementations are provided below.

- DITRAS [13]. DITRAS decouples temporal pattern generation from spatial location assignment by modeling temporal dynamics with a mobility diary generator and assigning these diaries to locations using the density-EPR model [14]. The diary generator employs a time-aware Markov transition matrix to estimate transition probabilities between "typical" locations (e.g., home) and non-typical locations, along with the corresponding durations of stays. Non-typical locations are then assigned to locations with the density-EPR mechanism, an extension of the classic EPR model that incorporates a relevance factor during exploration. We utilized the DITRAS implementation from scikit-mobility to align with our data format. Using the training dataset, we estimated the population-level Markov transition matrix and calibrated individual-specific density-EPR parameters. The relevance of locations was determined by their empirical visit frequencies. A temporal resolution of one hour was chosen to simulate 336 steps for test sequences, corresponding to 14 days of spatiotemporal mobility.

- TimeGeo [10]. Originally developed to generate high-resolution spatiotemporal mobility trajectories from sparse human digital traces, TimeGeo incorporates three individual-level parameters to characterize temporal movement dynamics: the weekly home-based tour number, the dwell rate, and the burst rate. These parameters capture time-dependent travel tendencies through a Markov decision process, distinguishing movement patterns based on an individual's location state. The framework employs a rank-based EPR model for spatial decisions, selecting exploration locations based on their proximity rank relative to the current position rather than absolute geographic distance. We estimated individual-specific temporal and EPR parameters using the training dataset. For each testing sequence, we designated the most frequently visited location as the home location and simulated 336 consecutive hourly movement steps, representing a 14-day mobility trajectory.

We assess the performance of these models using mobility metrics that capture activity-travel patterns and interactions. In addition to the metrics described in the main text, we introduce two supplementary metrics: unique location visits per day and activity start times. The evaluation results are presented in Supplementary Figure S2. To quantify model performance, we calculate the Wasserstein distance between the metric distributions derived from the generated activity events and those from the real data (Extended Data Table 2 and Supplementary Table S3). Metrics calculated from the source sequence are included as a strong baseline achievable by a mobility model; however, the variability in human activity participation must be considered, as source sequences vary in both the duration and timing of activity events.

MobilityGen consistently outperformed all other models in delineating individual activity-travel patterns and interactions, demonstrating a substantial margin of improvement. Furthermore, it provided descriptions that were as accurate as, or even more accurate than, those of the original source sequence.

Supplementary Note 4. Effectiveness of jointly learning activity attributes and movement-related contexts

We conduct ablation studies on MobilityGen to assess the contribution of incorporating activity attributes and contextual information. The activity attributes play dual roles in the simulation model: they provide additional data incorporated into the embedding space for joint mobility behavior learning (Equation 9) and impose further constraints on model optimization through the loss function (Equation 18). We start with a location model and progressively integrate time and travel mode data into the pipeline. All other training configurations are maintained to ensure a fair comparison. The results of the ablation experiments are summarized in Supplementary Table S4 and Supplementary Table S5, where model performance is evaluated using the Wasserstein distance between generated and real sequences across various mobility metrics. For each model variant, the simulation procedure is repeated three times with different random seeds, and the mean and standard deviation of the Wasserstein distances are reported. While the base model, which learns exclusively from historical location patterns, already exhibits strong performance compared to baseline mobility models (see performance numbers in Extended Data Table 1 and Supplementary Table S1), incorporating either time-use or travel mode information further improves its ability to replicate empirically observed mobility patterns. When all behavioral attributes are considered, the full model achieves the best performance across all metrics, with a slight decrease observed only in jump length. These results support our claim that jointly learning behavioral attributes facilitates a more realistic reproduction of behavior patterns.

Movement-related contexts, including location geometry and surrounding land-use functions, are integrated with activity attributes within the embedding space. We hypothesize that this integration enhances the ability of MobilityGen to characterize locations and is particularly useful for identifying novel locations for exploration. To evaluate this hypothesis, we compare the full model with a model variant that excludes contextual information. As described in the main text, the generated locations are classified into location sets based on their presence in the observed sequences: Set A includes locations common to both the observed and generated data, while Sets B and C contain locations unique to the observed and generated data, respectively. We then use the Wasserstein distance to quantify discrepancies in the distribution of visits to functional areas. We expect the model incorporating contextual information to generate more similar location preference patterns for unique locations (e.g., comparisons between Sets B and C) and more distinct patterns between shared and unique locations (e.g., comparisons between Sets A and C) than the model without contextual information. This hypothesis is supported by the Wasserstein distance comparisons in Supplementary Table S6, although the differences between models for Sets B and C patterns are not significant. This is likely due to the relatively low visit frequencies to these locations, resulting in large fluctuations in the visit patterns.

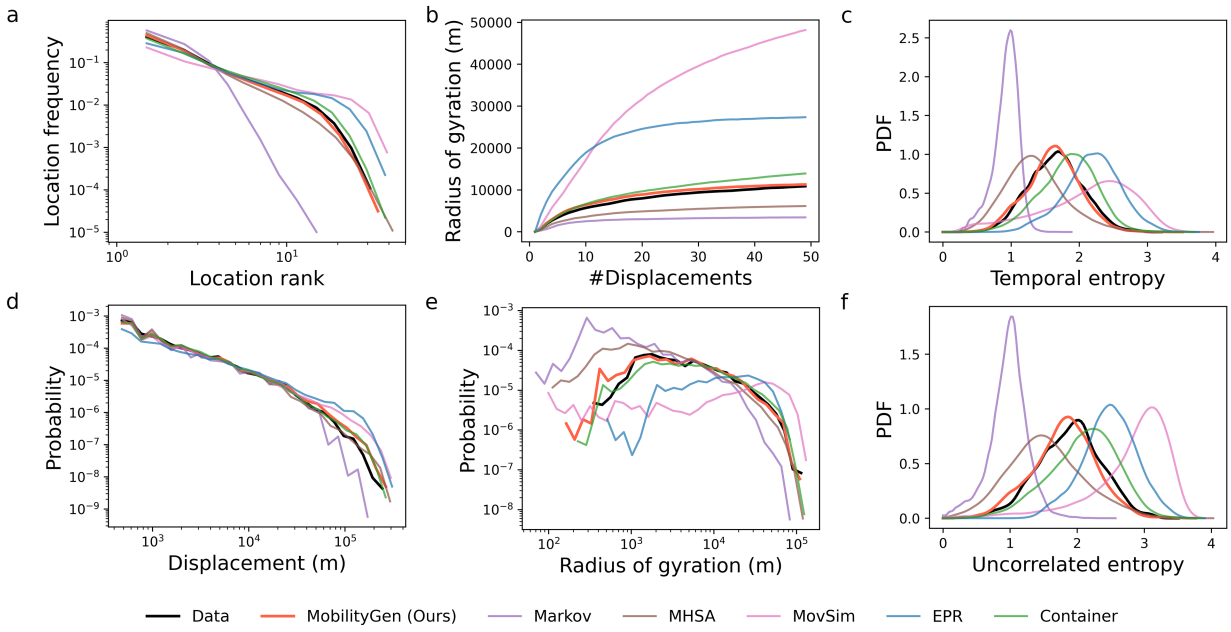

Figure S1: Evaluating microscopic mobility models with location metrics. **a**. Average visitation frequency versus location rank across individuals. **b**. Median individual radius of gyration plotted against the number of displacements. **c**. Distribution of temporal mobility entropy across individuals. **d** Distribution of displacements for the entire population. **e**. Distribution of radius of gyration among individuals. **f**. Distribution of uncorrelated mobility entropy across individuals. These metrics are compared between real data (black line) and traces generated by MobilityGen (red line), Markov (purple line), MHSA (brown line), MovSim (pink line), EPR (blue line), and container (green line).

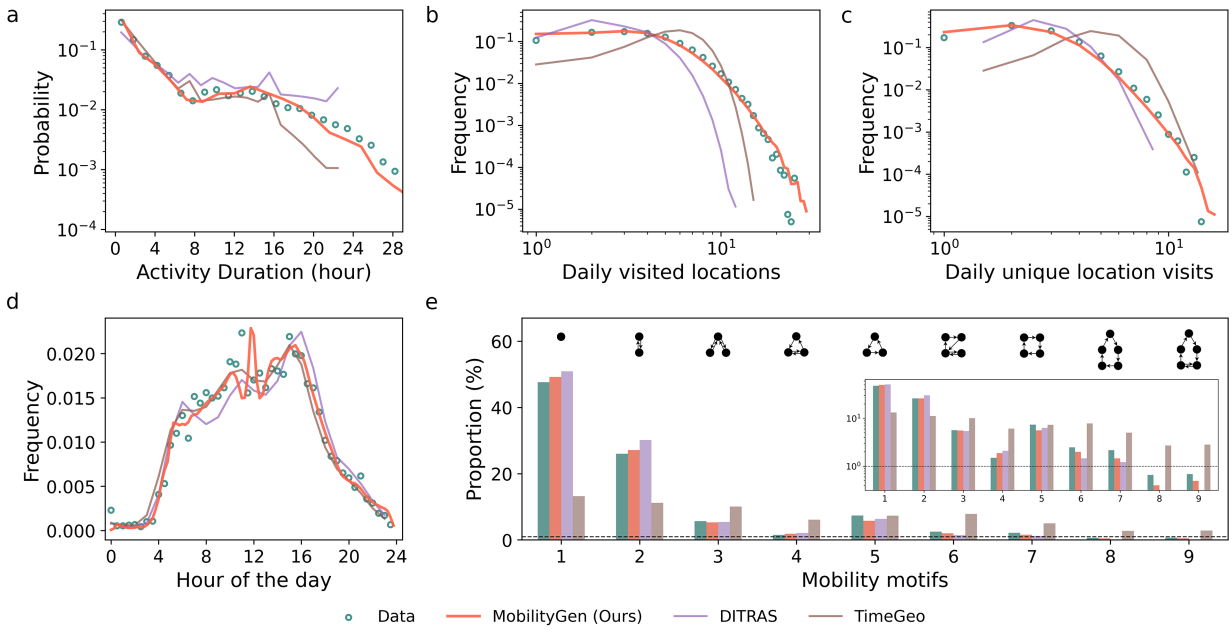

Figure S2: Generated events replicate individual activity-travel patterns and their interactions. **a**. Distribution of activity durations. **b**. Distribution of daily visited locations. **c**. Distribution of daily unique visited locations. **d**. Distribution of activity start times throughout the day. **e**. Distribution of mobility motifs patterns, with the dashed horizontal line marking 1% as a visual guide. The inset displays the same figure on a log scale. Metrics are compared between real data (green) and activity events generated by MobilityGen (red line), DITRAS (purple line), and TimeGeo (brown line).

Table S1: Performance of individual mobility models in simulating location visits, evaluated through additional metrics. We use the log-likelihood ratio to measure the likelihood that the real data was generated by each model relative to MobilityGen, with positive values indicating a stronger likelihood for MobilityGen (all $P < 10^{-3}$ except $P = 0.66$ for $P(f_{\Delta r})$ of MHSA). We additionally use the Wasserstein distance to assess the similarity between metric distributions for locations generated by each model and the real data, with higher values indicating greater dissimilarity. Results are presented for jump length ($f_{\Delta r}$), radius of gyration (f_{r_u}), and uncorrelated entropy (S_{unc}), with the best-performing metrics highlighted in bold.

	Log-likelihood			Wasserstein distance		
	$P(f_{\Delta r})$	$P(f_{r_u})$	$P(S_{\text{unc}})$	$f_{\Delta r}$	f_{r_u}	S_{unc}
EPR	166326	31364	74711	15882	12825	0.60
Container	-6568	946	2741	**1146**	3578	0.20
Markov	2697	16530	329477	2223	10102	0.91
MHSA	-115	4352	8355	1491	5137	0.35
MovSim	35384	79922	48838	8128	34096	0.92
MobilityGen (Ours)	-	-	-	2318	**575**	**0.08**

Table S2: Evaluating microscopic mobility models with language metrics. The BLEU score measures the similarity between the generated locations and the real data, with higher values indicating greater similarity. Distinct unigrams (dist-1) and diverse 4-grams (div-4) assess sequence diversity, where higher values indicate greater diversity.

	BLEU	dist-1	div-4
Markov	0.28	0.08	0.02
MHSA	0.24	0.21	0.29
MovSim	0.002	0.51	0.60
EPR	0.09	0.42	0.77
Container	0.19	0.31	0.57
MobilityGen (Ours)	0.26	0.23	0.34

Table S3: Benchmarking generated behavioral patterns and their interactions with additional metrics. We use the Wasserstein distance to assess the similarity between metric distributions for activity events generated by each model and the real data, with higher values indicating greater dissimilarity. Results are presented for unique daily visited locations ($f_{|k|}^{\text{day}}$) and activity start time (f_t), with the best-performing metrics highlighted in bold.

	$f_{\|k\|}^{\text{day}}$	f_t
DITRAS	0.37	1.49
TimeGeo	1.78	1.48
MobilityGen (Ours)	**0.22**	**0.52**

Table S4: Ablation studies on the effectiveness of jointly learning activity attributes using location metrics. We compare the performances of the full model with models that consider only location information (base) and progressively incorporate time (base + time use) and travel mode (base + travel mode). Wasserstein distances between the simulated and real traces are presented for the following metrics: number of visits per location (f_k), jump length ($f_{\Delta r}$), radius of gyration (f_{r_u}), temporal entropy (S_{temp}), and uncorrelated entropy (S_{unc}), with the best-performing metrics highlighted in bold.

	f_k	$f_{\Delta r}$	f_{r_u}	S_{temp}	S_{unc}
Base	0.649 $\pm$0.003	3871 $\pm$16	901 $\pm$21	0.110 $\pm$0.0004	0.216 $\pm$0.001
Base + time use	0.543 $\pm$0.002	**2280** $\pm$21	625 $\pm$19	0.101 $\pm$0.0005	0.144 $\pm$0.001
Base + travel mode	0.519 $\pm$0.002	3493 $\pm$15	804 $\pm$13	0.088 $\pm$0.0003	0.136 $\pm$0.001
Full model	**0.262** $\pm$0.004	2318 $\pm$14	**574** $\pm$17	**0.036** $\pm$0.001	**0.080** $\pm$0.001

Table S5: Ablation studies on the effectiveness of jointly learning activity attributes using activity behavior metrics. We compare the performance of the full model with models that consider only time (base + time use) or travel mode (base + travel mode). Wasserstein distances between the simulated and real traces are presented for the following metrics: activity duration (f_d), activity start time (f_t), daily visited locations (f_k^{day}), unique daily visited locations ($f_{|k|}^{\text{day}}$), mobility motifs (f_{motifs}), and travel mode (f_m), with the best-performing metrics highlighted in bold.

	f_d	f_t	f_k^{day}	$f_{\|k\|}^{\text{day}}$	f_{motifs}	f_m
Base + time use	0.904 $\pm$0.01	0.73 $\pm$0.01	0.447 $\pm$0.002	0.273 $\pm$0.001	0.233 $\pm$0.01	-
Base + travel mode	-	-	-	-	-	0.113 $\pm$0.001
Full model	**0.473** $\pm$0.001	**0.52** $\pm$0.01	**0.292** $\pm$0.01	**0.224** $\pm$0.002	**0.142** $\pm$0.004	**0.048** $\pm$0.002

Table S6: Ablation study on the impact of movement-related contexts. We compare the performance of a model without contextual information (MobilityGen w/o ctx) to the full model. Location sets are derived by comparing real and generated sequences, and discrepancies in the distribution of visits to functional areas are quantified using the Wasserstein distance. The distances include: real vs. generated distributions of Set A (d_A), Set C vs. Set B (d_B^C), Set C vs. real distribution of Set A ($d_{A_{observe}}^C$), as well as Set C vs. generated distribution of Set A ($d_{A_{generate}}^C$). Arrows indicate whether smaller or larger distances are preferred.

	$d_A \downarrow$	$d_B^C \downarrow$	$d_{A_{observe}}^C \uparrow$	$d_{A_{generate}}^C \uparrow$
MobilityGen w/o ctx	0.025 $\pm$0.001	0.153 $\pm$0.067	0.548 $\pm$0.001	0.545 $\pm$0.001
MobilityGen	0.009 $\pm$0.001	0.105 $\pm$0.006	0.560 $\pm$0.005	0.558 $\pm$0.003

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